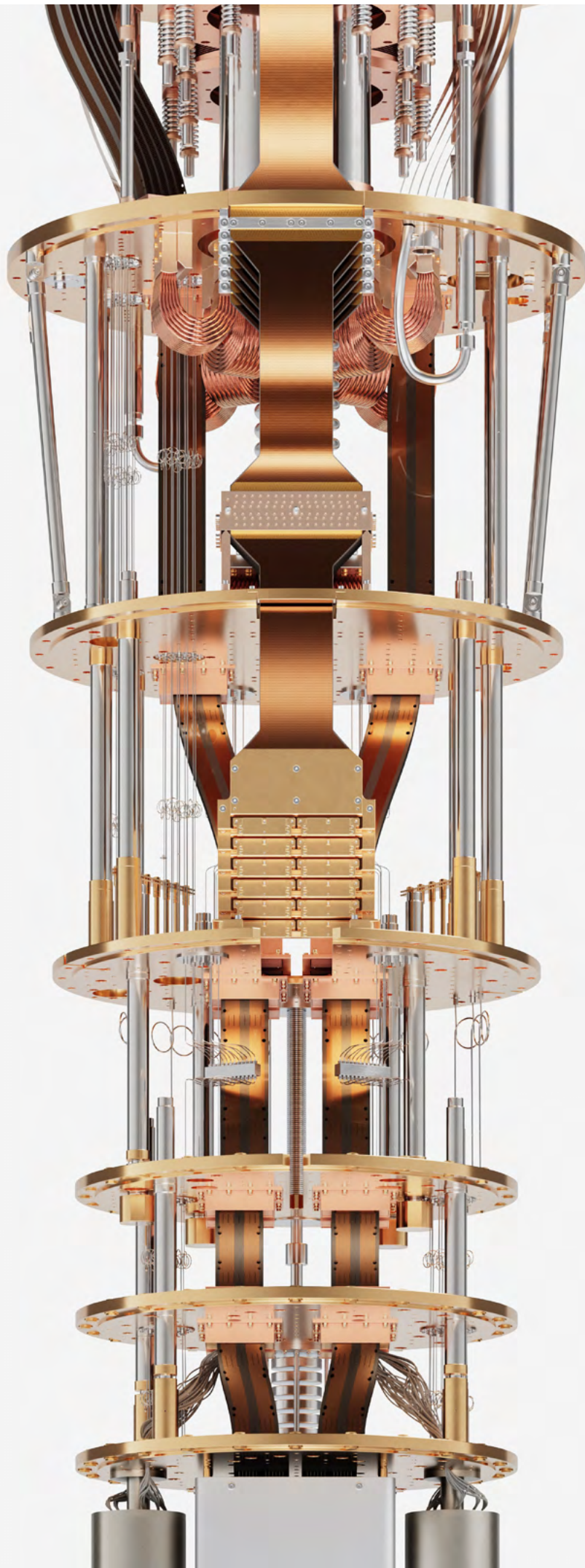
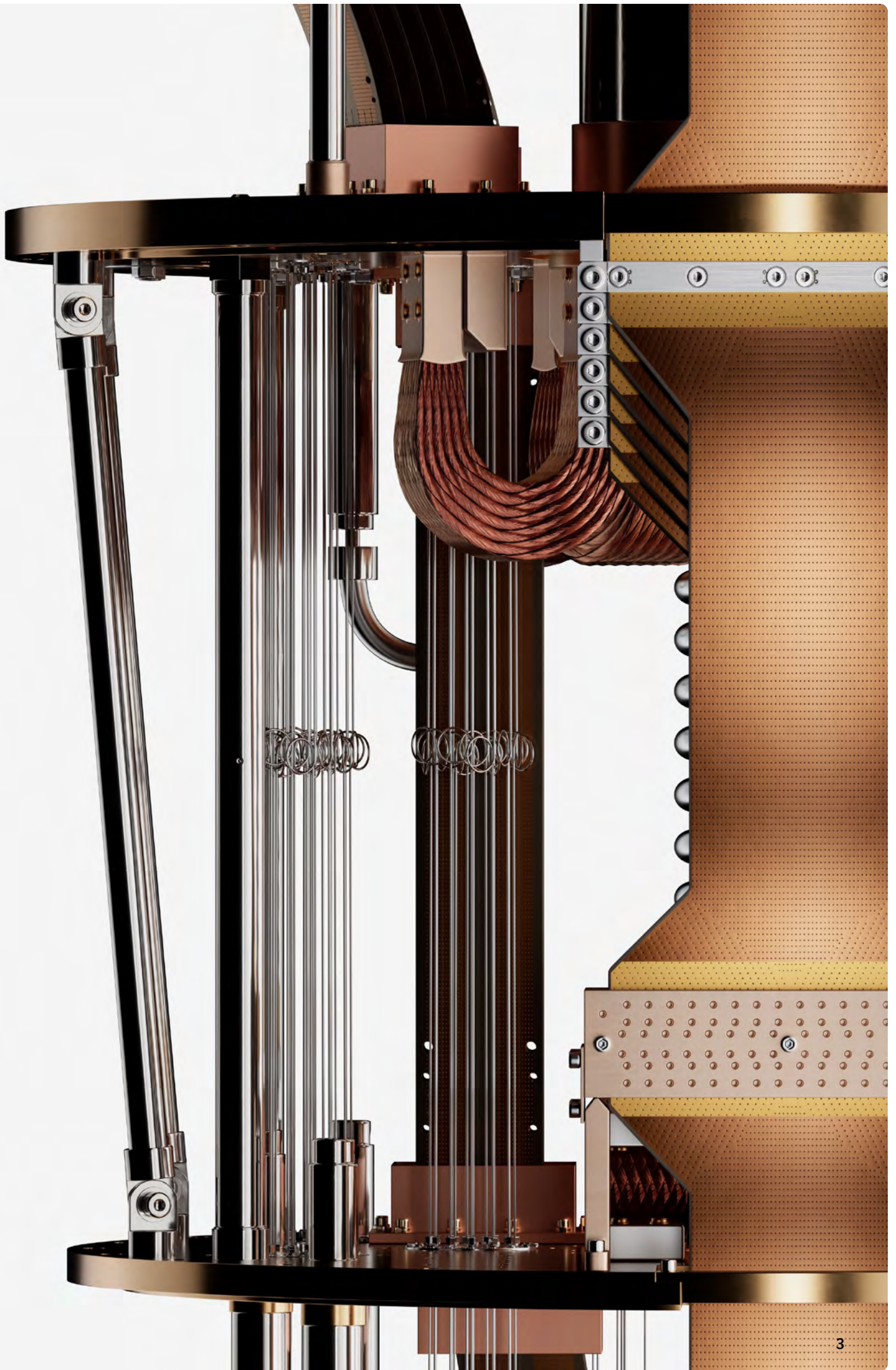


Journey to quantum advantage

Voices from the frontlines





Leading the way

Quantum computing stands at a pivotal moment. The science has moved decisively beyond theory: organizations are far along the path to quantum advantage, when quantum computers can run a computation more accurately, cheaply, or efficiently than any classical method.

Journey to Quantum Advantage captures the momentum steadily building around the world. Based on interviews with leaders from across industries—aerospace, finance, life sciences, energy, government, education, and more—it collects real-world stories underscoring quantum’s wide-ranging potential.

Complementing the IBM Institute for Business Value (IBM IBV)’s ongoing research on quantum readiness¹, this report celebrates the innovators and risk takers—those who seized frontier opportunities and are now advancing out front. Spurred by curiosity and visions of new capabilities, these leaders have overcome emergent complexities involving algorithms, hardware, and talent.

The organizations featured in this report didn’t avoid obstacles; their leaders adapted to them. They shifted from isolated experiments to place strategic bets. They refined the meaning of quantum advantage. Convinced that partnerships power innovation, these organizations invested in ecosystems connecting industry, government, university researchers, and other stakeholders. Many of these ecosystems didn’t exist a few years ago—a measure of growing maturity in the quantum space.

Taken together, these journeys illustrate what quantum readiness looks like. IBM IBV research finds quantum-ready organizations stand out not only for their technology. They seek accelerated innovation, solve intractable business problems, and future-proof their computing strategy. While the exact moment of quantum advantage will vary by use case and industry, these organizations all share something crucial in common: leaders who are building the future right now.

Strategic bets



02



Quantum journeys rarely follow a straight line, but all are guided by strategy. Across industries, leaders define long-term goals and make disciplined choices to achieve them. Some organizations focus on a few high-potential use cases, while others prioritize steady progress through broader exploration. Staying the course requires flexibility. Early optimism can be tested in the face of unclear return on investment or technical challenges. But what unites all organizations moving toward quantum advantage is strategic vision: the pursuit of innovation where it matters most, and a willingness to align capabilities and teams in pursuit of solving real-world problems.

Top 3 barriers to adoption¹

61% Inadequate quantum skills

56% Immature quantum technology

46% Unclear timelines of use cases

E.ON

Powering innovation

Year quantum journey began 2019	Industry Energy
Use cases Optimization, quantum machine learning, scenario modeling	



When E.ON began its quantum journey, excitement was high across the company. Early proofs of concepts suggested that quantum techniques could reshape key elements of Europe's energy systems, from grid balancing to forecasting renewables.

As Dr. Giorgio Cortiana points out, however, pursuing quantum advantage is not straightforward.

"Starting the quantum journey is relatively easy given the technology's disruptive potential," says Dr. Cortiana, Head of Quantum Computing at Germany-based E.ON, one of Europe's largest energy companies. "The real difficulty is clarifying when and how quantum's benefits will be realized."

The company's initial wave of exploratory pilots confirmed the underlying promise of quantum methods but also exposed their limitations: shallow circuits, limited qubit capacity, and hardware unable to outperform well-established classical methods. While the promise was real, hardware performance was not quite there yet.

“Leading-edge innovation demands *consistency* and *endurance*. Success typically comes from long-term commitment that allows capabilities, understanding, and opportunity to mature together.”



Dr. Giorgio Cortiana

Head of Quantum Computing,
E.ON

As experimentation moved toward more realistic, industrial problem formulations, the gap between available hardware and true advantage became increasingly clear. Error rates compounded. Circuit depth exceeded coherence windows and resource estimates for large-scale deployment escalated rapidly.

Business leaders understandably look for clarity on timelines and return, while technical teams operate at the frontier of what existing machines could support. The tension between research uncertainty and business expectations is not an anomaly, but a defining feature of applied research.

The resulting pressure led to a critical inflection point. E.ON recalibrated its internal roadmap to reflect the broader trajectory of quantum hardware development. With fault-tolerant quantum machines not expected before 2029, Dr. Cortiana and his team, together with E.ON's management, adjusted company expectations and reframed success accordingly. The search for near-term advantage—by means of increasingly hybrid algorithms and workflows—complemented long-term capability building.

E.ON's journey has tested assumptions. But it has also reinforced the company's conviction that quantum technologies will ultimately matter—albeit on a longer and less predictable timeline than initially hoped for.

“After the initial excitement, people start asking, ‘Should we wait a couple of years?’” Dr. Cortiana says. “That hesitation is natural. But what you build early only delivers value if you continue to evolve it over time. Leading-edge innovation demands consistency and endurance. Success typically comes not from isolated ideas, but from long-term commitment that allows capabilities, understanding, and opportunity to mature together.”

Dr. Cortiana's advice:

“Be clear about which part of the business quantum could provide an edge. If it's a small portion, consider concentrating on a specific use case.”

Yonsei University

Accelerating medical breakthroughs

Year quantum journey began 2024	Industry Higher education, medicine
Use case Drug discovery	



At South Korea's Yonsei University, medical research and quantum computing go hand in hand. Dr. Jae-Ho Cheong—a professor in the school's College of Medicine and Department of Quantum Information Science, and Director of the Yonsei Quantum Initiative—explains the university's strategic bets.

Why did Yonsei decide to launch its quantum initiative?

We believe a core part of the school's mission is to address difficult problems humanity is now faced with—an energy crisis, climate change, and especially global health burdens. A few years ago, university leaders were thinking carefully about how to fulfill that mission in the coming decade. They identified quantum as a crucial future-enabling technology, because current computational capabilities are not enough.

In the medical research space, current innovation and drug discovery practices struggle with speed and complexity. Progress is too slow and development is too costly. We believe advanced technologies—both AI and quantum computing—can fundamentally accelerate progress.

How do your areas of expertise—surgery, oncology, and technology—shape your quantum work?

I developed a gene based predictive test for cancer years ago, but I realized that prediction alone is not enough. We need more efficient, more accessible cancer therapies. When you look closely, many cancer types, especially clinically intractable subtypes, trace back to how energy is handled in mitochondria—specifically, electron transfer systems. That is a fundamentally quantum mechanical process. To truly understand and simulate it, classical computers are not sufficient. Quantum computers are much more natural tools to address this problem.

Yonsei is unique in having an on-site quantum computer. How does that influence your computing strategy?

We recognize the importance of hybrid quantum-classical workflows, which are essential for solving real-world medical problems. Our Eagle system is connected to Fugaku, the supercomputer at the RIKEN Center for Computational Science in Kobe, Japan. The physical distance is 800 kilometers [497 miles] but the latency period is just 20 milliseconds.

This quantum-centric high performance computing platform has led us to valuable cross-disciplinary collaborations with RIKEN scientists—chemists and physicists who rarely engage in life sciences research collaborations. Currently, Yonsei and RIKEN researchers as a team tackle one of the most formidable challenges in quantum chemistry: determining the electronic structure of Fe-S clusters in mitochondria with unprecedented accuracy

What does quantum advantage mean to you and your team, in practical terms?

From an end user perspective, quantum advantage does not need to be dramatic to be meaningful. The drug discovery process typically takes about 15 years. If quantum computing helps us reduce that time by, say, 10% or 20%, that changes lives. But quantum computing can also enable new methods that are impossible with classical computing. For example, we could potentially see better classification of cancer patients into more homogenous clinical sub-segments for better treatment matching.

Beyond research, how is Yonsei building long term quantum capabilities?

We made two big strategic decisions to lead the field in South Korea and the Pacific region. First, we launched a graduate degree program in quantum information science. We now have 20 Ph.D. students. Second, we decided to hire more faculty members capable of doing quantum research and education.

After we purchased and began operating our quantum computer, the question naturally arose: How could we nurture the next-generation workforce? We're now answering that question in our research facilities.

Dr. Cheong's advice:

“Start by asking the right questions to clearly define the problem you intend to solve.”

“The drug discovery process typically takes about 15 years. If quantum computing helps us reduce that time by, say, 10% or 20%, *that changes lives.*”



Dr. Jae-Ho Cheong

Director of the Yonsei Quantum Initiative,
Yonsei University

Bosch

Taking the harder path

Year quantum journey began 2017	Industry Automotive
Use cases Materials simulation for e-drives, fuel cells, electrolyzers, and advanced sensors	



Global engineering and technology company Bosch has had more success than most in hitting quantum computing milestones. The company successfully developed quantum algorithms and established a materials simulation workflow that, at the time, utilized the largest number of qubits applied to a real-world solid-state materials problem.

“At the beginning of our journey, we achieved significant progress,” says Dr. Thomas Eckl, Chief Expert at Bosch. He points to demonstrations on early, noisy quantum devices as a major technical milestone.

That momentum, however, met a familiar industry-wide obstacle: scaling. As Bosch attempted to move from proof of concept to larger, more realistic problem sizes,

the limitations hit. Noise levels remained too high and qubit connectivity too limited. Results increasingly fell short of expectations.

This set the stage for a choice. Bosch could continue pushing the early, noisy applications it had success with, or pivot to preparing for a fault tolerant future without having hardware available today. The company chose the harder path: moving beyond early wins toward a new roadmap. Its quantum program is now centered on developing robust scalable algorithms for the fault-tolerant regime, with a focus on resource estimation, workflow integration, and understanding algorithmic performance on real quantum hardware.

The adjustment has been more than technical. Internally, Bosch has had to manage expectations and sustain long term commitment amid rising economic pressures and scrutiny of quantum investments. “We began with enthusiasm and a broad vision,” Dr. Eckl says. “Now the focus has shifted to measurable outcomes: where will quantum deliver real value, and when will the ROI materialize?”

To sustain momentum, Bosch has increasingly focused on use cases with shorter feedback cycles and clearer near-term impact. While materials simulation remains a central long-term strategic investment, the company has broadened its efforts toward engineering applications such as optimization and partial differential equation solvers, where advances could accelerate design iterations and deliver business value faster. Demonstrating measurable progress on real-world problems step by step, Dr. Eckl argues, is critical for maintaining organizational engagement and long-term readiness.

“If we can demonstrate usefulness for relevant applications,” he says, “that builds confidence that quantum will ultimately solve problems we can’t address today.”



Dr. Eckl’s advice:

“Identify the computational bottlenecks limiting your organization’s ability to solve critical problems. Then evaluate whether existing or emerging quantum algorithms could overcome those limitations.”

“We began with enthusiasm and a broad vision. Now the focus has shifted to measurable outcomes: where will quantum deliver real value, and when will the ROI materialize?”

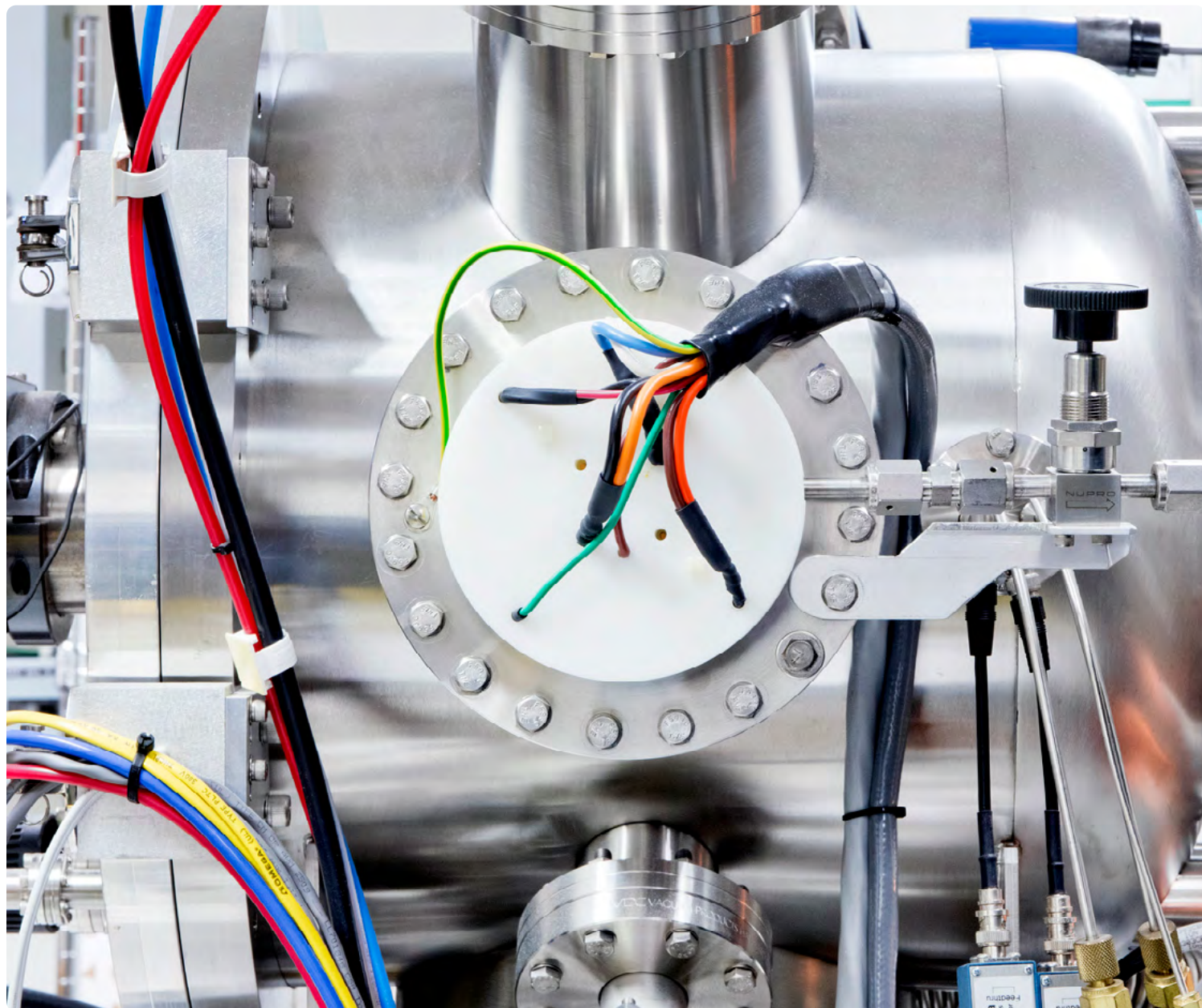
Dr. Thomas Eckl
Chief Expert,
Bosch

Chicago State University

Closing skills gaps



Year quantum journey began 2018	Industry Higher education
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Chicago State University (CSU) views quantum as both a platform for innovation and an engine of workforce development. As the city’s growing quantum ecosystem—which encompasses research universities, national labs, industry partners, Chicago Public Schools and others—gains steam, CSU President Zaldwaynaka “Z” Scott explains how her school is creating accessible pathways from classrooms to careers.

How did Chicago State University become a force in the city’s quantum ecosystem?

For nearly a decade, our scientists have been exploring quantum and how it integrates with other research areas. As the state of Illinois began to build a stronger tech ecosystem—especially around quantum and microelectronics—we saw a

clear alignment with Chicago State’s interests and mission. What really sharpened the strategy was recognizing the need for skilled workers in this emerging field. We started to look at how CSU could be a catalyst for inclusive innovation, given our location less than five miles from the planned Illinois Quantum Microelectronics Park (IQMP) on Chicago’s South Side and our status as the only federally designated four-year Predominantly Black Institution in Illinois.

How does CSU’s quantum strategy support skills development?

Our goal isn’t just to do advanced science. It’s to make sure communities understand the opportunity and are able to participate in it.

“When you coordinate intentionally—across institutions, sectors, and neighborhoods—
you can grow both science and communities together.”



President Zaldwaynaka Scott
President,
Chicago State University

Closing skills gaps requires intentional planning from the start. That means designing educational pathways that span elementary through high school, certificates, undergraduate study and master's degree programs, and research experiences. We're looking at it all as a way for us to engage.

What does that approach look like in practice?

We've partnered with the University of Chicago on quantum summer camps for high school students. We've brought Chicago Public Schools science teachers to our campus to give them training, tools, and materials to teach quantum concepts in their high-school classrooms. We offer entry-level microelectronics courses on our campus with the goal of preparing students for entry level jobs and real economic opportunity. That's just a few examples.

How is CSU balancing near term job readiness with long term quantum futures?

We've been very proactive. We are the first in Illinois to offer Illinois Board of Higher Education-certified quantum minor and quantum certificate programs so students can supplement their majors with relevant skills. We already have hundreds of students interested, and our first certificate graduates are entering the workforce now and making an impact. For instance, one graduate will begin a joint research project with Harvard and another will begin a research position at Pfizer. There is of course no way to offer students complete knowledge in such a fast-changing space.

We try to give students a strong foundation to take into jobs, where they can continue to grow and keep up depending on the field they enter.

How is the university supporting longer-term research and collaborations that push quantum forward?

We established the CSU Quantum Education, Science & Technology Center so this work has a permanent institutional home. We've built a cross disciplinary team at the center, because we know quantum won't live only in STEM, it will touch many disciplines. We hope this center helps to grow our collaborative relationship with Argonne National Laboratory. We're also a major educational partner in IQMP, a 128-acre research hub that will bring national labs, universities, government, and businesses together.

What does success look like 10 years from now?

Success is graduating students with quantum credentials that prepare them to work in that space. It's having upgraded laboratories where students are learning alongside experienced researchers to get the campus experience they deserve. It's seeing community members confident they aren't locked out of opportunity. And it's proving that when you coordinate intentionally—across institutions, sectors, and neighborhoods—you can grow both science and communities together.

President Scott's advice:

“Be intentional about who should be at the table. Make sure that your invitation list has a broad section of the community present.”

Basque Quantum (BasQ)

Forging long-term impact

Year quantum journey began
2016

Industry
Government

Use cases
Financial services optimization, biomedical



The Basque Country may be one of Spain's smaller regions, but its quantum profile looms large. Over the last 10 years, the Basque Government has built a cross-sector ecosystem with the goal of positioning the autonomous region as a global quantum player. Basque Country Vice-Minister of Science and Innovation Dr. Adolfo Morais and Basque Quantum (BasQ) Scientific Director Dr. Francisco Javier Aizpurua explain why and how.

How did the Basque Country land on quantum computing as a strategic priority?

Vice-Minister Adolfo Morais: The Basque Country is small in size and population, a reality that forces strategic choices. Around 10 years ago, the government launched a long term flagship research and impact program designed to identify areas that would truly matter for the future of the region. Quantum science and technology was

one of those areas, and our interest in it evolved into a strategic initiative for quantum, called BasQ. The ambition was to concentrate resources, talent, and policy, and link them to impact. We also wanted autonomy: access to advanced technology, the ability to develop our own workforce, and a mature ecosystem rather than isolated projects.

Once the government made the deliberate decision to explore the impacts of quantum science on industry and society, quantum computing emerged as a capability with the potential to transform multiple sectors.

Dr. Javier Aizpurua: And there was an important parallel evolution. On one hand, policymakers began supporting a focused strategy around quantum technologies. On the other hand, researchers and technologists deepened their work aligned to that vision.

Which sectors and organizations are part of the BasQ ecosystem?

Dr. Aizpurua: It's a collaborative ecosystem that includes universities, public research centers, applied mathematics institutes, startups, established industrial companies, and public sector organizations. We see strong engagement from sectors such as banking, advanced manufacturing, energy, and biomedical research. The key has been alignment: administration, science, and industry moving together rather than in isolation.

Vice-Minister Morais: From the start, collaboration between public and private actors was essential. This shared vision has allowed the ecosystem to grow coherently rather than fragment.

Has BasQ's goal and scope evolved over time?

Dr. Aizpurua: The core ambition has remained the same: to become an international reference in quantum technologies, particularly quantum computing. What has evolved is focus. Initially, the vision was broad and exploratory. Today, we are much more precise about where we can achieve excellence. At the same time, the scope has expanded—we are working across more challenges, more algorithms, and more applications. But always with a sharper definition of where we can truly lead.

Which research areas and use cases have emerged?

Dr. Aizpurua: On the scientific side, there is strong activity in electronic structure calculations, partial differential equations solving, and high energy physics, especially the simulation of gauge fields. On the applied side, we see promising work in optimization—particularly in financial services—as well as biomedical research and accelerated AI. The spectrum runs from fundamental science to highly practical optimization problems.

What lessons does the Basque Country's journey offer to other regions considering similar investments?

Vice-Minister Morais: A shared, long term vision is essential. Administration, researchers, and industry must advance together with a clear understanding of strengths and limitations. Quantum is a major bet. Public investment must be justified by long term value. For us, that alignment has been the foundation of the ecosystem's success.

Vice-Minister Morais' advice:

“Have a common vision encompassing policymakers, researchers, and industry, and be clear about your strengths and weaknesses.”

“From the start,
collaboration between
public and private
actors was essential.”



Dr. Adolfo Morais

Vice-Minister of Science and Innovation,
Basque Country

“The key has been
alignment: administration,
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Dr. Javier Aizpurua

Scientific Director,
Basque Quantum

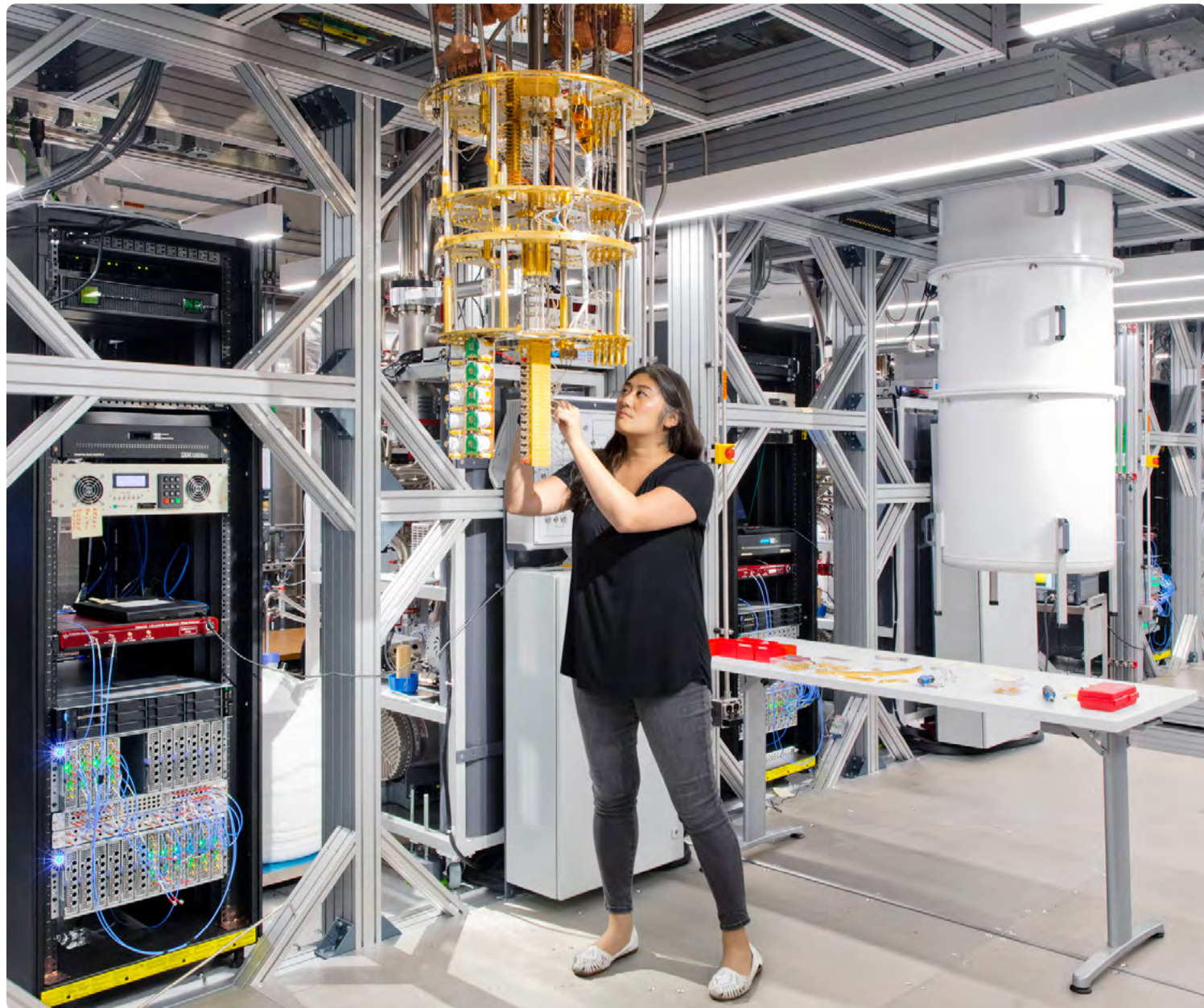
Tokyo Electron Limited (TEL)

Assembling expert teams



(by Dr. Akihisa Sekiguchi)

Year quantum journey began 2018	Industry Electronics
Use cases Equipment configuration and semiconductor process optimization, nanometer scale and macroscopic scale fluid simulations	



We first began exploring quantum computing at Tokyo Electron Limited (TEL) more than five years ago. Quantum was a technology worth learning about, but not yet ready for structured investment. That changed when we realized that meaningful progress would require education, shared learning, and long-term partnerships rather than isolated experiments.

With partners like Q-STAR [Japan-based Quantum Strategic industry Alliance for Revolution], we refined expectations, conducted joint application research, and identified real problems that classical systems struggle to solve. One of the most important lessons was the power of diverse expertise versus narrow specialization. Success requires teams that combine deep domain knowledge, from data science to classical computing to quantum implementation. No single organization naturally possesses all of them.

We were able to assemble multidisciplinary teams to work through real semiconductor challenges. Semiconductor manufacturing is inherently complex—it spans materials science, process physics, optimization, and data. We needed access to talent that would be difficult to secure independently.



Partnerships also matter because uncertainty is high. Hardware architectures are evolving, timelines are shifting, and benchmarks continue to change. By navigating that uncertainty with partners, you learn more, faster. Sharing insights about mistakes and iterating together makes that cycle more effective and more efficient.

Looking ahead, I firmly believe that quantum computing will only reach its potential through ecosystems of like-minded organizations. No single entity can address the technical, ethical, security, and societal implications alone. International collaboration, shared industry guidelines, and coordinated education efforts will all be necessary.

For TEL, quantum fits naturally with how we deliver value: faster innovation, better technology, and deeper insight into the physics and processes our customers care about. Building partnerships has allowed us to move closer to that vision. As quantum advantage emerges, it will belong not to isolated pioneers, but to those who invested in building ecosystems that power collaboration and innovation.

Dr. Sekiguchi's advice for beginning the quantum journey:

“Remember to pack domain experts, data scientists, HPC simulation algorithm experts, and quantum hardware and software specialists in your suitcase.”



“Quantum advantage will belong not to isolated pioneers, but to those who invested in building ecosystems that power collaboration and innovation.”

Dr. Akihisa Sekiguchi

Senior VP and GM of US R&D, Fellow,
Tokyo Electron Limited (TEL)

Republic of Singapore

Creating a quantum hub

Year quantum journey began 1998	Industry Government
Use cases Grid and investment portfolio optimization, solar panel and data center predictive maintenance, secure communications, logistics and supply chain optimization	



Singapore is committed to cultivating a thriving quantum ecosystem that brings government, academic, and private-sector organizations together to build readiness. Sustained government funding has helped establish flagship institutions—hosted by the National University of Singapore—which has produced highly-cited research and multiple industry spin offs.

“We have invested steadily over two decades,” says Choy Yong Cong, Director of International Affairs and Quantum Policy and Strategy at Singapore’s Ministry of Digital Development and Information. “This has positioned us well to capture the current trajectory of quantum tech development and seize opportunities.”

The latest phase of Singapore’s journey is the National Quantum Strategy, launched to strengthen capabilities across research, engineering, talent, translation, and partnerships. Its implementation is led by the National Quantum Office and supported by a network of government agencies working together to support ecosystem development, enterprise growth, and real-world quantum deployment. Agencies include the Ministry of Digital Development and Information (MDDI), the Economic Development Board (EDB), the Infocomm Media Development Authority (IMDA), the National Research Foundation (NRF), Enterprise Singapore, and SGInnovate.

Dr. Clifton Phua, Director of Labs at IMDA’s BizTech Group, sees a flywheel effect. “Real use cases create demand, demand attracts capability, capability drives investment, and investment strengthens the quantum computing ecosystem.”

A multi-pronged strategy

Singapore has invested more than SGD700 million in quantum technologies since 2002, reflecting a sustained national commitment to the field. Singapore's National Quantum Strategy, launched in 2024, builds on this strong foundation, committing an additional SGD300 million over five years. The strategy takes a more coordinated approach to strengthening its quantum ecosystem, spanning research, engineering, talent, and collaboration. This whole-of-government effort spans complementary agencies that together support research and deployment pathways, help local firms scale, and attract foreign quantum companies to the city-state. The strategy has four focus areas:

Scientific excellence. Singapore has built a strong base in frontier quantum research through institutions like the Centre for Quantum Technologies, which is a global leader in areas including quantum communications and sensing.

Engineering capabilities. National platforms such as the Quantum Engineering Programme and the National Quantum Processor Initiative have strengthened Singapore's ability to translate research into deployable quantum technologies.

Partnerships. International and industry partnerships form a core part of Singapore's approach to quantum, helping local researchers and companies access infrastructure, co-develop use cases, and plug into global quantum value chains.

Talent and research-to-industry translation. Singapore is investing in scholarships, industry-linked programs, and national platforms to develop talent and translate research into real-world applications and commercial opportunities.

This emphasis on strengthening the quantum ecosystem extends into academia. At the National University of Singapore, quantum computing research has never been treated as a standalone academic pursuit. Researchers work closely alongside public agencies to explore credible use cases.

According to George Loh, Associate Vice President (Strategic Partnership) at the university, success has depended less on the number of partnerships than on their depth: "You need partners who are willing to invest and experiment for the long term, not those looking for short-term returns."

Loh points to openness and public commitments as essential tools for aligning stakeholders and maintaining momentum in an uncertain field. "What I appreciate is transparency—having a clear roadmap that shows where we are going and how progress will be measured," he says.

Ultimately, Singapore's strategy is about preparedness in the face of uncertainty. By building national platforms and cultivating dense networks of collaboration, Singapore has established an ecosystem through which quantum advantage can emerge. "As a government, we can't afford to wait and see," Choy says. "We have to prepare for quantum now."

George Loh's advice:

"Seriously commit and build a core team first—even three or four people can be enough to begin."

“You need partners who are willing to invest and experiment for the long term, not those looking for short-term returns.”



George Loh

Associate Vice President (Strategic Partnership),
National University of Singapore

“As a government, we can’t afford to wait and see. We have to prepare for quantum now.”

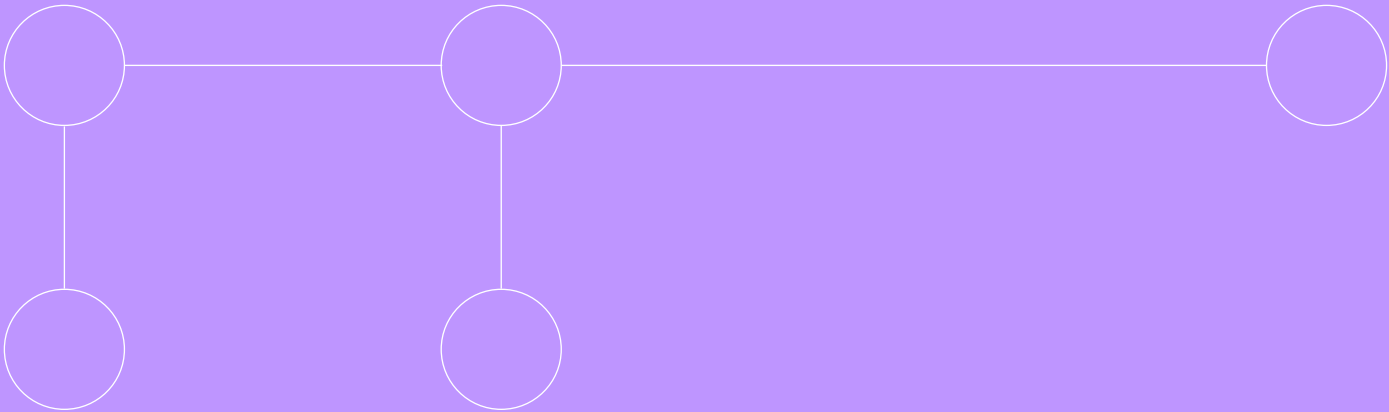


Choy Yong Cong

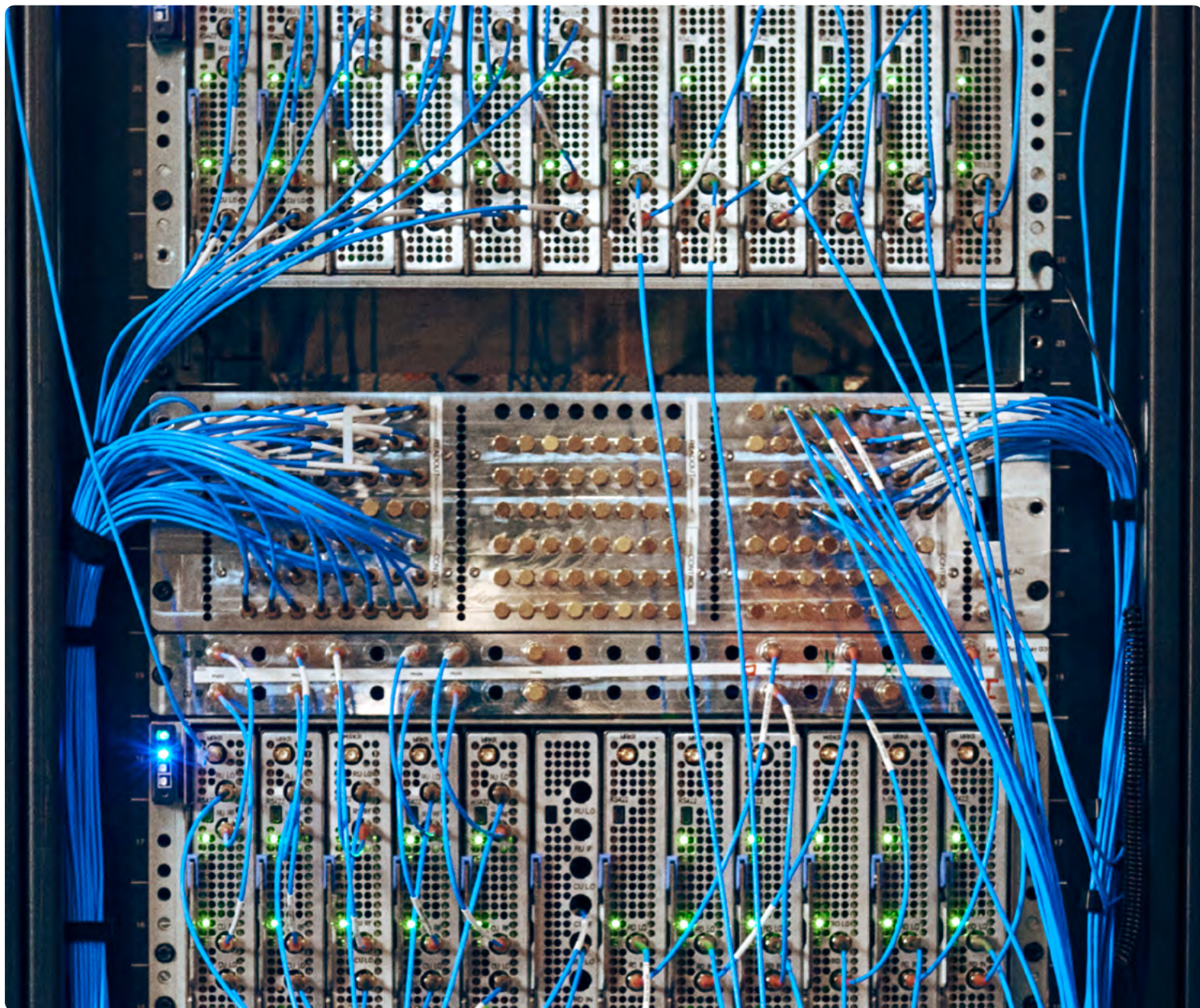
Director of International Affairs and Quantum Policy and Strategy,
Ministry of Digital Development and Information

Bradesco

Banking on partnerships



Year quantum journey began 2019	Industry Banking
Use cases Portfolio optimization, risk analysis, macroeconomic variable forecasting	



Dr. Jonas Rossetti—who leads the quantum research team at Bradesco, one of Brazil’s largest financial institutions—explains his bank’s quest for quantum use cases, from portfolio optimization to forecasting.

How do you infuse quantum into Bradesco’s innovation strategy?

I like to think of our quantum journey as supported by four pillars: research, experiments, training, and partnerships. Together, these pillars allow us to achieve quantum readiness.

Research helps us understand how the technology is evolving—hardware, software, opportunities, and limitations. Experiments connect that knowledge to real financial problems by

bringing together quantum specialists and domain experts from across the bank. Training builds our internal capabilities. We offer many training paths within Bradesco to deepen knowledge and skills, such as quantum computing learning programs and courses.

But we can’t progress alone—we need to be an active part of the quantum community. International conferences, local events, and joint research help us build credibility, attract talent, and position Bradesco as an innovation-driven bank.

What role do partnerships play?

For us, partnerships are fundamental because they allow us to accelerate learning and reduce uncertainty. Industry partners give us access to hardware and tools and educational materials, while academic partners help us develop methods that are not yet ready for production. Our partnership with the University of São Paulo, for example, helps us test hypotheses, explore algorithms, and exchange perspectives. From a broader strategic perspective, external collaborations help us stay close to the frontier without over-committing internally before the technology matures.

How does outside collaboration help you build internal business value?

For me, quantum advantage only exists if it creates value that outweighs the cost of investment. It's not enough to say an algorithm is better. We have to show that it improves operational efficiency or decision-making in a way that justifies the investment. That's why I prefer a diversified portfolio of use cases—portfolio optimization, risk analysis, forecasting—rather than betting everything on one problem. Collaboration helps us learn across these areas without locking ourselves into a single, risky path too early.

What are the key lessons from your quantum journey?

Quantum readiness doesn't come from hype or isolated pilot projects. Advantage won't be just about hardware or algorithms. Success flows from people, partnerships, and deepening understanding over time. Stay the course so that when the technology is ready, the organization is ready too.

Dr. Rossetti's advice:

“You don't need a large team to start. But you do need a dedicated one and a clear intention to grow capability and knowledge, both internally and through partnerships.”

“External collaborations help us stay close to the frontier *without overcommitting* internally before the technology matures.”



Dr. Jonatas Rossetti

R&D Specialist,
Bradesco

Conclusion

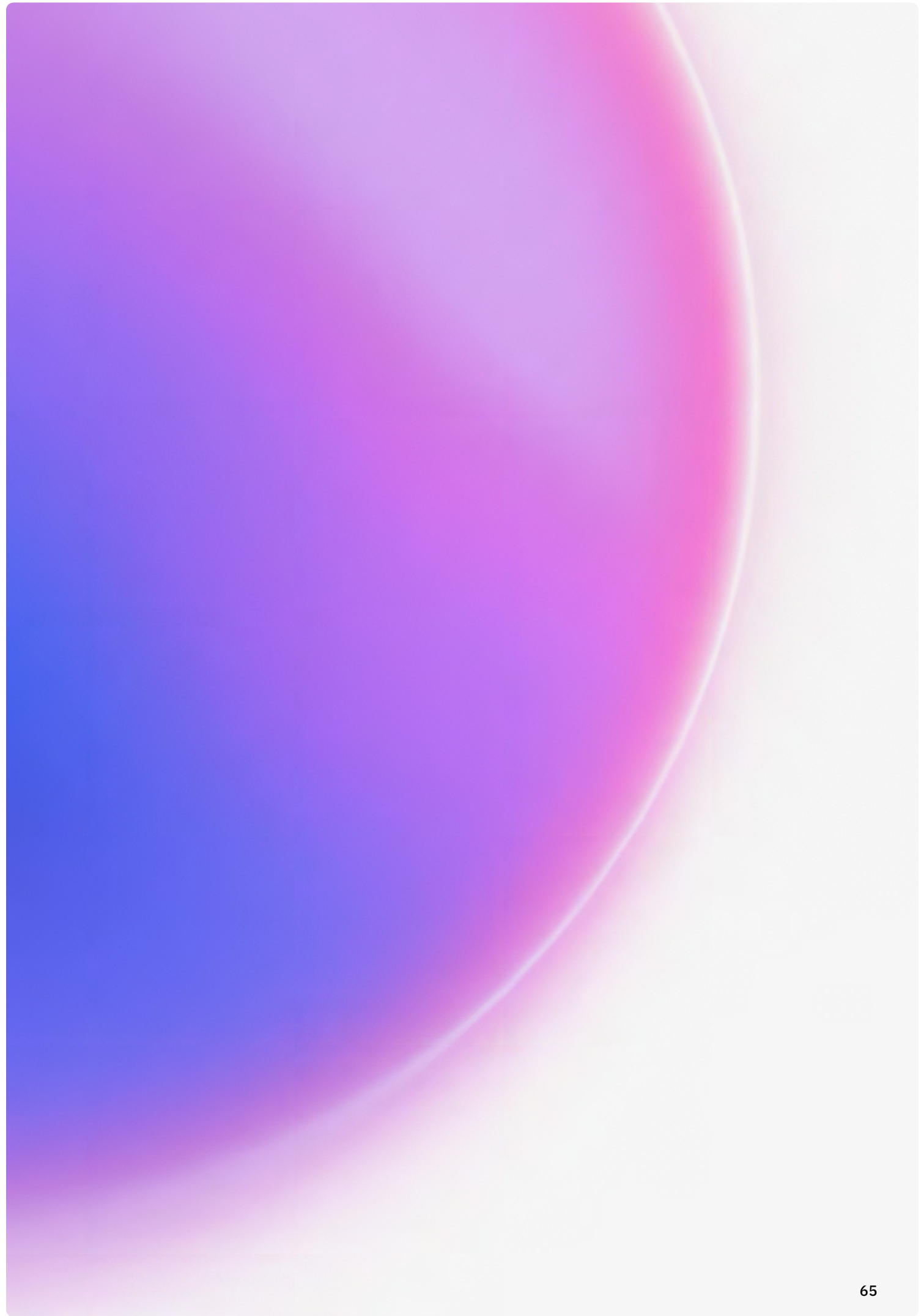
The journey ahead

Quantum computing is undoubtedly a transformative force, one that demands foresight and collaboration. Leading organizations aren't waiting for fault-tolerant systems to arrive—they're taking action now, homing in on specific, uniquely valuable use cases. Very often, progress is accelerating through ecosystems that connect businesses, government, academia, hardware providers, and research institutions.

Living at the frontier of innovation, first-movers are showing how quantum systems can augment—not replace—existing technologies, as hybrid architectures are deployed. Quantum and AI will work together to analyze complex problems that lie well beyond today's computational limits.

What once was decades in the future is happening now. Across industries and around the world, organizations are building the conditions necessary to seize a new competitive edge. The most successful will likely be those that invest early in building foundational quantum capabilities to avoid being left behind.

33% increase in quantum investment when quantum and AI are framed as complementary.¹



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Raja leads the global delivery practice to drive useful quantum computing applications for enterprises across all industries. He brings delivery excellence leadership to accelerate progress toward quantum business advantage. Raja has deep global experience in driving enterprise strategy and large business and technology transformation programs, and in delivering valuable quantum applications for enterprise use cases.

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A growth-minded entrepreneur and inspirational executive at IBM, Heather leads a world-class team of global quantum industry and technical experts working with pioneering enterprises that are evaluating and adopting quantum technology for commercial impact. With 30 years at IBM, she has deep expertise at the intersection of business and emerging technology, combined with proven experience helping organizations harness technology to unlock value.

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Veena leads quantum computing research at the IBM Institute for Business Value. Previously, she held leadership positions in semiconductor product development, technology strategy, and management. Her research findings have influenced clients in industries ranging from electronics to government to financial services. Veena is a frequent speaker at major conferences in North America, Asia, and Europe. She serves as a Senior Quantum Ambassador for IBM.

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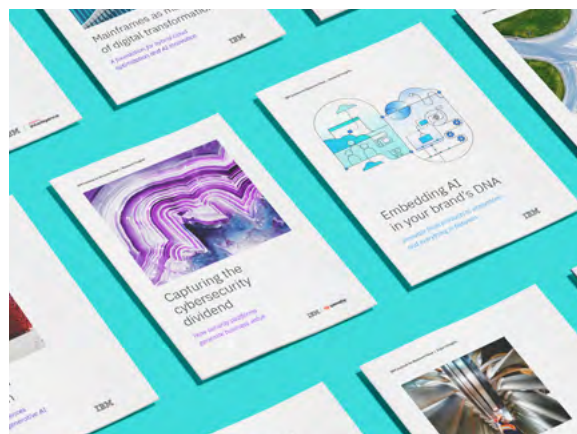
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